

Recap.

① Set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$

Union $\cup$

Intersection $\cap$

Complement $A^c = \Omega \setminus A$

De Morgan's Law: $(A \cup B)^c = A^c \cap B^c$

$$(A \cap B)^c = A^c \cup B^c$$

Power set $2^\Omega = \mathcal{P}(\Omega) = \{A \mid A \subseteq \Omega\}$ (All subsets)

② Samples: $\Omega = \{a, b, c\}$.

Events: $A_1 = \emptyset$ $A_2 = \{a\}$ $A_3 = \{b, c\}$... $A_n = \{a, b, c\}$, subsets of Ω .

Event systems: subsets of $\{\emptyset, \{a\}, \{b\}, \dots, \{a, b, c\}\}$, in total $\mathcal{P}(\mathcal{P}(\Omega))$.

③ Ω : Full space of samples "Probability of A": $P(A) = \frac{|A|}{|\Omega|}$.

A: Samples in the event

④ "Under the condition B, the probability of A": $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

Exercises.

Ex 1. Full space: $\Omega = \{(1,1), (1,2), \dots, (6,6)\}$, $|\Omega| = 36$.

(i) Event A: sum is 7.

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

$A = \{(1,6), (2,5), \dots\}$, $|A| = 6$.

(ii) Event B: sum ≥ 10 .

$$P(B) = \frac{|B|}{|\Omega|} = \frac{1}{6}.$$

$B = \{(5,5), (4,6), \dots\}$, $|B| = 6$.

(iii) Event C: at least one die shows a 6.

$\bar{C}$: none of both is 6.

$$P(C) = 1 - P(\bar{C}) = 1 - \frac{|\bar{C}|}{|\Omega|} = \frac{11}{36}.$$

$C^c = \Omega \setminus C = \{(1,1), (1,2), \dots\}$, $|C^c| = 25$.

Ex 2.

$$(i) \frac{3}{5} \times \frac{2}{4} = \frac{1}{10}.$$

$$(ii) P(\text{exactly one is red}) = P((R, B) \text{ or } (B, R)) = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{3}{4} = \frac{3}{5}.$$

$$(iii) P(\text{second blue}) = P((R, B) \text{ or } (B, B)) = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} = \frac{2}{5}.$$

$$(iv) P(\text{first red} \mid \text{second blue}) = \frac{P((R, B))}{P(\text{second is blue})} = \frac{\frac{3}{5} \times \frac{2}{4}}{\frac{2}{5}} = \frac{3}{4}.$$

$$(v) P(\text{exactly one blue} \mid \text{least one blue}) = \frac{P((R, B) \text{ or } (B, R))}{P((R, B) \text{ or } (B, R) \text{ or } (B, B))} = \frac{P((R, B) \text{ or } (B, R))}{1 - P((R, R))} = \frac{6}{7}.$$

Ex 3. Hint: Choose a from b , there are $\binom{b}{a} = \binom{b}{b-a}$ outcomes.

$$(i) P(\text{all distinct}) = \frac{10}{10} \times \frac{9}{10} \times \frac{8}{10} \times \frac{7}{10} = \frac{63}{125}.$$

$$(ii) P(\text{at least one repeat}) = 1 - P(\text{all distinct}) = 1 - \frac{63}{125} = \frac{62}{125}.$$

$$(iii) P(\text{exactly one pair}) = \frac{10 \times 9 \times 8 \times \binom{4}{2}}{10^4} = \frac{54}{125}.$$

$$(iv) P(\text{palindrome}) = \frac{10 \times 10}{10^4} = \frac{1}{100}.$$

Ex 4.

$$(i) \quad 2\% \times 10\,000 = 200 \text{ submissions have bug.}$$

$$200 \times 90\% = 180 \text{ will be flagged.}$$

$$(ii) \quad 10\,000 \times 98\% = 9800 \text{ submissions don't have bug.}$$

$$9800 \times 5\% = 490 \text{ will be flagged.}$$

$$(iii) \quad \frac{180}{180 + 490} = \frac{18}{67}.$$

Ex 5.

$$(i) \quad 3 \text{ Aces and in total } 51 \text{ cards left, } \frac{3}{51}.$$

$$(ii) \quad \text{Known: at least one is Ace.}$$

$\Rightarrow$ All the situations with no Ace shown up should not be considered.

$$|\Omega'| = \binom{52}{2} - \binom{48}{2}$$

$$P(\text{both are Aces}) = \frac{\binom{4}{2}}{\binom{52}{2} - \binom{48}{2}} = \frac{1}{33}.$$